Problem Set 3

Instructor solutions

Clock arithmetic, Fermat's little theorem, and repeating continued fractions

Week 3 | 100 points + 10 optional challenge points

Teaching note. Estimated working time is 2–3 hours, excluding the challenge. Problems 1–5 use only congruences, inverses, and Fermat's little theorem; the Chinese remainder theorem, Euler's theorem, and the notion of order are not assumed. Problems 6–8 use the integer-state algorithm, Galois's criterion, and the period-reversal corollary from the second continued-fraction lecture. All floors are decided by integer bounds on $\sqrt{D}$; decimal approximations should not receive proof credit. Subpart scores appear in parentheses. Avoid repeatedly penalizing a carried arithmetic error.

Instructions. Work exactly: no decimal approximations. When you need the floor of a surd, justify it with integer bounds such as $2 < \sqrt{7} < 3$. Justify every proof or counterexample. You may use Bézout's identity, Euclid's lemma in the form "if $n \mid cm$ and $\gcd(n, c) = 1$, then $n \mid m$," and the fact from Problem Set 2 that coprime divisors of c have a product dividing c.

Facts from lecture. *Fermat's little theorem:* if p is prime, then $a^p \equiv a \pmod{p}$ for every integer a, and $a^{p-1} \equiv 1 \pmod{p}$ when $p \nmid a$. *Integer states:* if $x_n = (P_n + \sqrt{D})/Q_n$ with $Q_n \mid D - P_n^2$ and $a_n = \lfloor x_n \rfloor$, then

$$P_{n+1} = a_n Q_n - P_n, \qquad Q_{n+1} = \frac{D - P_{n+1}^2}{Q_n}, \qquad x_{n+1} = \frac{P_{n+1} + \sqrt{D}}{Q_{n+1}}.$$

Galois: a quadratic surd x has a purely periodic continued fraction exactly when it is *reduced*: $x > 1$ and $-1 < x' < 0$, where x' is the conjugate. *Reversal:* if $x = [\overline{b_0, \ldots, b_{\ell-1}}]$, then $-1/x' = [\overline{b_{\ell-1}, \ldots, b_0}]$.

I. Congruences and inverses

1. Warming up the clock

[10 points]

(a) Find the least nonnegative residue of -100 modulo 13. (2)

(b) Find the last two digits of $99 \cdot 101 \cdot 103$ without multiplying the numbers out. (2)

(c) Prove that every square is congruent to 0 or 1 modulo 4. Deduce that no integer congruent to 3 modulo 4 is a sum of two squares. Is 2027 a sum of two squares? (6)

Solution

(a) $-100 = -8 \cdot 13 + 4$, so the residue is 4.

(b) Modulo 100 the factors are -1, 1, and 3. Their product is $-3 \equiv 97$, so the last two digits are 97. (The product is positive, so its last two digits are its residue modulo 100.)

(c) Every integer is congruent to $0, 1, 2$, or 3 modulo 4, and congruences may be squared. The squares of these residues are $0, 1, 4, 9 \equiv 0, 1, 0, 1$. Hence a sum of two squares is congruent to $0 + 0$, $0 + 1$, or $1 + 1$, that is, to 0, 1, or 2; the residue 3 never occurs. Since $2027 = 4 \cdot 506 + 3$, it is not a sum of two squares.

2. Inverses by Euclid

[14 points]

(a) Use the extended Euclidean algorithm to find the inverse of 23 modulo 89. Check your answer by multiplication. (5)

(b) Solve $23x \equiv 5 \pmod{89}$. (2)

(c) Let $n \geq 2$. Prove that an inverse modulo n, when it exists, is unique modulo n: if $ab \equiv 1$ and $ac \equiv 1 \pmod{n}$, then $b \equiv c \pmod{n}$. (3)

(d) List the units modulo 15, and give the inverse of each. Which units are their own inverses? (4)

Solution

(a) The divisions are

$$89 = 3 \cdot 23 + 20, \qquad 23 = 1 \cdot 20 + 3, \qquad 20 = 6 \cdot 3 + 2, \qquad 3 = 1 \cdot 2 + 1.$$

Working backward,

$$1 = 3 - 2 = 7 \cdot 3 - 20 = 7 \cdot 23 - 8 \cdot 20 = 31 \cdot 23 - 8 \cdot 89.$$

Hence $23^{-1} \equiv 31 \pmod{89}$. Check: $23 \cdot 31 = 713 = 8 \cdot 89 + 1$.

(b) Multiply by 31: $x \equiv 155 \equiv 66 \pmod{89}$. Check: $23 \cdot 66 = 1518 = 17 \cdot 89 + 5$. The solution is unique modulo 89 because multiplication by the unit 23 can be undone.

(c) Using only associativity and commutativity of multiplication of residues,

$$b \equiv b \cdot 1 \equiv b(ac) \equiv (ab)c \equiv 1 \cdot c \equiv c \pmod{n}.$$

(d) The units are the residues coprime to 15: $1, 2, 4, 7, 8, 11, 13, 14$. The inverse pairs are $2 \cdot 8 = 16$, $7 \cdot 13 = 91 = 6 \cdot 15 + 1$, and the self-inverse units are $1, 4, 11, 14$ ($4^2 = 16$, $11^2 = 121 = 8 \cdot 15 + 1$, $14 \equiv -1$). Note that there are four square roots of 1 here; compare Problem 3(b).

3. Cancellation, square roots of one, and Wilson's theorem [14 points]

(a) Prove the cancellation law: if $ac \equiv bc \pmod{n}$ and $\gcd(c, n) = 1$, then $a \equiv b \pmod{n}$. Give an example showing that the conclusion can fail when $\gcd(c, n) > 1$. (5)

(b) Let p be prime. Prove that $a^2 \equiv 1 \pmod{p}$ forces $a \equiv \pm 1 \pmod{p}$. Where does your proof break for the modulus 8? (4)

(c) Pair each of the residues $2, 3, \ldots, 9$ with its inverse modulo 11, and use the pairing to compute $10!$ modulo 11 with almost no arithmetic. Then prove *Wilson's theorem*: $(p-1)! \equiv -1 \pmod{p}$ for every prime p. (5)

Solution

(a) The hypothesis says $n \mid c(a-b)$. Since $\gcd(n, c) = 1$, Euclid's lemma gives $n \mid a-b$. (Alternatively, multiply both sides by an inverse of c.) (3) For a failure, $2 \cdot 1 \equiv 2 \cdot 4 \pmod{6}$ but $1 \not\equiv 4 \pmod{6}$. (2)

(b) The hypothesis says $p \mid a^2 - 1 = (a-1)(a+1)$. A prime dividing a product divides a factor, so $a \equiv 1$ or $a \equiv -1 \pmod{p}$. (3) Modulo 8 we have $3^2 \equiv 1$ although $3 \not\equiv \pm 1$: here $8 \mid 2 \cdot 4$ without dividing either factor, so the step "divides a product, hence divides a factor" requires a prime. (1)

(c) Modulo 11: $2 \cdot 6 = 12$, $3 \cdot 4 = 12$, $5 \cdot 9 = 45 = 4 \cdot 11 + 1$, and $7 \cdot 8 = 56 = 5 \cdot 11 + 1$. Therefore

$$10! \equiv 1 \cdot (2 \cdot 6)(3 \cdot 4)(5 \cdot 9)(7 \cdot 8) \cdot 10 \equiv 10 \equiv -1 \pmod{11}. \quad (2)$$

In general, the case $p = 2$ is $1! \equiv -1 \pmod{2}$. Let p be odd. Each of $1, \ldots, p-1$ is coprime to p, so it has an inverse in the same list, unique by Problem 2(c); and the inverse of the inverse of a is a. By (b), a is its own inverse exactly when $a \equiv \pm 1$, that is, for $a = 1$ and $a = p - 1$ only. The remaining residues $2, \ldots, p-2$ therefore split into pairs $\{a, a^{-1}\}$ of distinct elements, each pair with product 1. Hence $(p-1)! \equiv 1 \cdot (p-1) \equiv -1 \pmod{p}$. (3)

II. Fermat's little theorem

4. Large powers made small [12 points]

(a) Compute $3^{1000} \pmod{13}$. (3)

(b) Compute $5^{2026} \pmod{23}$. (3)

(c) Let p be prime and $p \nmid a$. Explain why a^{p-2} is an inverse of a modulo p. Use this to find the inverse of 7 modulo 19 by repeated squaring, and check your answer by multiplication. (6)

Solution

(a) By Fermat, $3^{12} \equiv 1 \pmod{13}$. Since $1000 = 12 \cdot 83 + 4$,

$$3^{1000} \equiv (3^{12})^{83} \cdot 3^4 \equiv 81 \equiv 3 \pmod{13}.$$

(Even faster: $3^3 = 27 \equiv 1$ and $1000 = 3 \cdot 333 + 1$.)

(b) By Fermat, $5^{22} \equiv 1 \pmod{23}$, and $2026 = 22 \cdot 92 + 2$. Hence $5^{2026} \equiv 5^2 = 25 \equiv 2 \pmod{23}$.

(c) $a \cdot a^{p-2} = a^{p-1} \equiv 1 \pmod{p}$ by Fermat. (2) For $a = 7$, $p = 19$ we need $7^{17} = 7^{16} \cdot 7$. Squaring repeatedly,

$$7^2 = 49 \equiv 11, \qquad 7^4 \equiv 121 \equiv 7, \qquad 7^8 \equiv 11, \qquad 7^{16} \equiv 7 \pmod{19},$$

so $7^{17} \equiv 7 \cdot 7 \equiv 11$. Check: $7 \cdot 11 = 77 = 4 \cdot 19 + 1$. (4)

5. Fermat as a proof tool, and a warning [14 points]

(a) Prove that $30 \mid n^5 - n$ for every integer n. (5)

(b) Let p be a prime other than 2 and 5. Prove that p divides infinitely many of the numbers 9, 99, 999, 9999, ... (4)

(c) Verify that $2^{10} = 3 \cdot 341 + 1$, and deduce that $2^{340} \equiv 1 \pmod{341}$. Yet $341 = 11 \cdot 31$ is not prime. Does this contradict Fermat's theorem? Finally, use Fermat modulo 31 to show that $3^{340} \not\equiv 1 \pmod{341}$, and explain how this congruence alone proves that 341 is composite. (5)

Solution

(a) Fermat with $p = 5$ gives $n^5 \equiv n \pmod{5}$ directly. Modulo 3, Fermat gives $n^3 \equiv n$, so $n^5 = n^3 \cdot n^2 \equiv n \cdot n^2 = n^3 \equiv n$. Modulo 2, $n^2 \equiv n$ gives $n^5 = (n^2)^2 n \equiv n^2 \cdot n \equiv n \cdot n \equiv n$. Thus 2, 3, and 5 each divide $n^5 - n$. Since $\gcd(2, 3) = 1$, their product 6 divides $n^5 - n$; since $\gcd(6, 5) = 1$, so does 30. (Award full credit for the factorization $n^5 - n = (n-1)n(n+1)(n^2+1)$ with a complete case analysis modulo 5.)

(b) The number with m nines is $10^m - 1$. Since $p \nmid 10$, Fermat gives $10^{p-1} \equiv 1 \pmod{p}$, hence $10^{k(p-1)} \equiv 1$ for every $k \geq 1$. Thus p divides $10^{k(p-1)} - 1$, the number with $k(p-1)$ nines, for every $k \geq 1$; these are infinitely many different numbers.

(c) $3 \cdot 341 = 1023 = 2^{10} - 1$. Hence $2^{10} \equiv 1$ and $2^{340} = (2^{10})^{34} \equiv 1 \pmod{341}$. (2) There is no contradiction: Fermat's theorem says that primes satisfy the congruence; it does not say that only primes do. The *converse* of Fermat's theorem is false. (1) For base 3: by Fermat, $3^{30} \equiv 1 \pmod{31}$, and $340 = 30 \cdot 11 + 10$, so $3^{340} \equiv 3^{10} \pmod{31}$. Now $3^3 = 27 \equiv -4$, so $3^9 \equiv -64 \equiv -2$ and $3^{10} \equiv -6 \equiv 25 \pmod{31}$. If 3^{340} were 1 modulo 341, it would also be 1 modulo the divisor 31; it is not. Finally, if 341 were prime, Fermat (with $341 \nmid 3$) would force $3^{340} \equiv 1 \pmod{341}$. The failure of this congruence proves compositeness without exhibiting a factor. (2) (In fact $3^{340} \equiv 56 \pmod{341}$.)

III. Continued fractions of quadratic surds

6. Running the integer-state algorithm [14 points]

(a) Make a table of n, P_n, Q_n, x_n, a_n for $x_0 = \sqrt{7}$, continuing until a state (P_n, Q_n) repeats. State the continued fraction of $\sqrt{7}$. (6)

(b) Let $x = (1+\sqrt{2})/3$. Explain why $(P_0, Q_0) = (1, 3)$ with $D = 2$ is not a legal starting state. Show that $x = (3+\sqrt{18})/9$ is legal, and run the algorithm to find the continued fraction of x. (6)

(c) Which of the complete quotients $x_0, x_1, x_2, \ldots$ in (b) are reduced? Compare with where the period begins. (2)

Solution

(a) Every floor follows from $2 < \sqrt{7} < 3$.

n	P_n	Q_n	x_n	a_n
0	0	1	$\sqrt{7}$	2
1	2	3	$(2+\sqrt{7})/3$	1
2	1	2	$(1+\sqrt{7})/2$	1
3	1	3	$(1+\sqrt{7})/3$	1
4	2	1	$2+\sqrt{7}$	4
5	2	3	$(2+\sqrt{7})/3$	1

For instance, $4 < 2+\sqrt{7} < 5$ gives $4/3 < x_1 < 5/3$, so $a_1 = 1$; then $P_2 = 1\cdot 3 - 2 = 1$ and $Q_2 = (7-1)/3 = 2$. Since $x_5 = x_1$,

$$\sqrt{7} = [2; \overline{1, 1, 1, 4}].$$

(b) A legal state needs $Q_0 \mid D - P_0^2$, but $3 \nmid 2 - 1 = 1$; the recurrence would produce the non-integer Q_1. Multiplying numerator and denominator by 3 gives $x = (3+\sqrt{18})/9$, and $9 \mid 18 - 9 = 9$. (2) Using $4 < \sqrt{18} < 5$:

n	P_n	Q_n	x_n	a_n
0	3	9	$(3+\sqrt{18})/9$	0
1	-3	1	$-3+\sqrt{18}$	1
2	4	2	$(4+\sqrt{18})/2$	4
3	4	1	$4+\sqrt{18}$	8
4	4	2	$(4+\sqrt{18})/2$	4

Here $7/9 < x_0 < 8/9$, $1 < x_1 < 2$, $4 < x_2 < 9/2$, and $8 < x_3 < 9$. Since $x_4 = x_2$,

$$\frac{1+\sqrt{2}}{3} = [0; 1, \overline{4, 8}]. \quad (4)$$

(c) $x_0 < 1$ is not reduced. $x_1 > 1$, but its conjugate $-3-\sqrt{18}$ is below -1, so x_1 is not reduced. The conjugate of x_2 is $(4-\sqrt{18})/2$, which lies in $(-1/2, 0)$, and the conjugate of x_3 is $4-\sqrt{18} \in (-1, 0)$; both exceed 1, so both are reduced, as is every later x_n. The reduced complete quotients are exactly the ones inside the period, as Galois's theorem requires: x_n is purely periodic precisely for $n \geq 2$.

7. From a period back to a surd [12 points]

(a) Let $x = [\overline{3, 1}]$. Find a quadratic equation with integer coefficients satisfied by x, and write x in the form $(P + \sqrt{D})/Q$. Confirm directly that x is reduced. (5)

(b) Use the reversal corollary to find $t = [\overline{1, 3}]$ in closed form, and then find $y = [2; \overline{1, 3}]$. (4)

(c) Your answers should satisfy $y = x - 1$. Explain this by looking only at the digits. Is y purely periodic? Reconcile your answer with Galois's theorem. (3)

Solution

(a) Removing one period leaves x again:

$$x = 3 + \frac{1}{1 + \dfrac{1}{x}} = 3 + \frac{x}{x+1} = \frac{4x+3}{x+1}.$$

(In matrix form, $D(3)D(1) = \left(\begin{smallmatrix} 4 & 3 \\ 1 & 1 \end{smallmatrix}\right)$.) Hence $x^2 - 3x - 3 = 0$. Since $x > 0$,

$$x = \frac{3 + \sqrt{21}}{2}.$$

From $4 < \sqrt{21} < 5$ we get $7/2 < x < 4$, so $x > 1$, and $x' = (3 - \sqrt{21})/2$ lies between -1 and $-1/2$. Thus x is reduced.

(b) By the corollary,

$$t = -\frac{1}{x'} = \frac{2}{\sqrt{21} - 3} = \frac{2(\sqrt{21} + 3)}{12} = \frac{3 + \sqrt{21}}{6}.$$

Then

$$y = 2 + \frac{1}{t} = 2 + \frac{6}{3 + \sqrt{21}} = 2 + \frac{6(\sqrt{21} - 3)}{12} = \frac{1 + \sqrt{21}}{2}.$$

(c) Writing out the period, $x = [3; 1, 3, 1, 3, \ldots] = [3; \overline{1, 3}]$, whereas $y = [2; \overline{1, 3}]$. The two expansions have the same tail t and first digits differing by one: $x = 3 + 1/t$ and $y = 2 + 1/t$. (1) The number y is not purely periodic: the conjugate $y' = (1 - \sqrt{21})/2$ satisfies $y' < (1 - 4)/2 < -1$, so y is not reduced, and Galois's theorem excludes pure periodicity. (2) (Contrast the lecture example $[2; \overline{1, 2}] = [\overline{2, 1}]$, where the displayed prefix was not a genuine preperiod. Here it is genuine: a purely periodic expansion of y with first digit 2 would have to be $[\overline{2, 1, 3, \ldots}]$, and uniqueness of the expansion would force the digit 2 to recur.)

8. Galois's test and two infinite families [10 points]

(a) Without computing any continued fractions, decide which of

$$2 + \sqrt{7}, \qquad \frac{3 + \sqrt{5}}{2}, \qquad \frac{2 + \sqrt{7}}{3}, \qquad \frac{1 + \sqrt{7}}{4}$$

have purely periodic continued fractions. Then read off the expansions of the purely periodic ones from your table in Problem 6(a). (4)

(b) Let $n \geq 1$. Prove that $\sqrt{n^2 + 1} = [n; \overline{2n}]$. (3)

(c) Let $n \geq 1$. Find and prove a formula for the continued fraction of $\sqrt{n^2 + 2}$. (3)

Solution

(a) Use $2 < \sqrt{7} < 3$ and $2 < \sqrt{5} < 3$.

x	$x > 1$?	x'	Reduced?
$2+\sqrt{7}$	yes	$2-\sqrt{7} \in (-1, 0)$	yes
$(3+\sqrt{5})/2$	yes	$(3-\sqrt{5})/2 > 0$	no
$(2+\sqrt{7})/3$	yes, $x > 4/3$	$(2-\sqrt{7})/3 \in (-1/3, 0)$	yes
$(1+\sqrt{7})/4$	no, $x < 1$		no

The reduced ones are the complete quotients x_4 and x_1 of $\sqrt{7}$. Reading the digits from those rows onward,

$$2+\sqrt{7} = [\overline{4, 1, 1, 1}], \qquad \frac{2+\sqrt{7}}{3} = [\overline{1, 1, 1, 4}].$$

(b) Put $D = n^2 + 1$. From $n^2 < D < (n+1)^2$ we get $n < \sqrt{D} < n+1$, so $a_0 = n$ and

$$x_1 = \frac{1}{\sqrt{D} - n} = \frac{\sqrt{D} + n}{D - n^2} = \sqrt{D} + n.$$

Since $2n < x_1 < 2n+1$, $a_1 = 2n$ and $x_2 = 1/(\sqrt{D} - n) = x_1$. Every later digit is therefore $2n$.

(c) The formula is $\sqrt{n^2+2} = [n; \overline{n, 2n}]$. Put $D = n^2 + 2$. Since $n \geq 1$ gives $n^2 < D < n^2 + 2n + 1$, again $n < \sqrt{D} < n+1$ and $a_0 = n$. The states are

$$(P_1, Q_1) = (n, 2), \quad x_1 = \frac{n+\sqrt{D}}{2} \in \left(n,\, n+\tfrac{1}{2}\right), \quad a_1 = n;$$

$$(P_2, Q_2) = \left(2n - n,\ \frac{D-n^2}{2}\right) = (n, 1), \quad x_2 = n + \sqrt{D} \in (2n, 2n+1), \quad a_2 = 2n;$$

and then $(P_3, Q_3) = (2n - n, (D-n^2)/1) = (n, 2) = (P_1, Q_1)$. The states repeat with period two. For $n = 1$ this recovers $\sqrt{3} = [1; \overline{1, 2}]$, and $n = 2$ gives $\sqrt{6} = [2; \overline{2, 4}]$.

Optional challenge: the period of $\sqrt{D}$ is a palindrome

[10 extra points]

Let D be a positive nonsquare integer and $a_0 = \lfloor \sqrt{D} \rfloor$. Prove that

$$\sqrt{D} = [a_0; \overline{a_1, a_2, \ldots, a_{\ell-1}, 2a_0}] \qquad \text{with} \qquad a_i = a_{\ell-i} \quad (1 \leq i \leq \ell - 1).$$

Compare with $\sqrt{7}$ in Problem 6. *Hint:* apply Galois's theorem and the reversal corollary to $x = a_0 + \sqrt{D}$. You may use the fact that an irrational number has only one simple continued fraction expansion.

Solution

Let $x = a_0 + \sqrt{D}$. Then $x > 1$, and $x' = a_0 - \sqrt{D} \in (-1, 0)$ because $a_0 < \sqrt{D} < a_0 + 1$. By Galois's theorem x is purely periodic, and its first digit is $\lfloor x \rfloor = 2a_0$. Let ℓ be a period length and write

$$x = [\overline{2a_0, a_1, \ldots, a_{\ell-1}}].$$

Subtracting a_0 changes only the first digit, so $\sqrt{D} = [a_0; a_1, \ldots, a_{\ell-1}, 2a_0, a_1, \ldots] = [a_0; \overline{a_1, \ldots, a_{\ell-1}, 2a_0}]$, which is the first claim.

Now compute $-1/x'$ in two ways. Directly, $-1/x' = 1/(\sqrt{D} - a_0)$, which is the first complete quotient of x; deleting the first digit of x gives

$$-\frac{1}{x'} = [\overline{a_1, a_2, \ldots, a_{\ell-1}, 2a_0}].$$

Solution (continued)

By the reversal corollary applied to x,

$$-\frac{1}{x'} = \overline{[a_{\ell-1}, \ldots, a_2, a_1, 2a_0]}.$$

By uniqueness of the expansion, the two digit sequences agree term by term: $a_i = a_{\ell-i}$ for $1 \le i \le \ell - 1$. For $\sqrt{7} = [2; \overline{1, 1, 1, 4}]$ the palindrome is $1, 1, 1$ and the last digit is $4 = 2 \cdot 2$.
Suggested credit: x is reduced, hence purely periodic with first digit $2a_0$ (3); shape of the expansion of $\sqrt{D}$ (2); two expressions for $-1/x'$ (3); comparison by uniqueness (2).